

Subject-Wise Syllabus

CS221 — Unit-wise Topic Coverage (Theory & Practical)

Advanced Machine Learning

Course Code: 24CSH-338

Unit 1

Advanced Supervised Algorithms

- Ridge, Lasso, and Group Lasso Regression
- Variants of SVM and Support Vector Regression (SVR)
- One-Class SVM and Least Squares SVM (LS-SVM)
- SVM in high-dimensional feature spaces

Unsupervised Learning Algorithms

- Introduction to clustering with Need and Role of clustering
- Hierarchical Clustering: AGNES and DIANA
- Topology-based Clustering: Self- Organizing Map
- Probabilistic Sequence/Latent Methods: Gaussian Mixture Models

Advanced Ensemble Learning Algorithms

- Ensemble Learning Basics and Combination Schemes with Voting
- Weighted Averaging and Voting Strategies with Majority Voting, Hard and Soft Voting
- Cross-validation, Types and Significance
- Bagging Method: Random Forest
- Boosting Method: Adaboost and Gradient Boosting
- Boosting Method: XGBoost
- Stacking and blending methods

Unit 2

Probabilistic, Distributed and Federated Learning Architectures

- Introduction to Probabilistic Graphical Model
- Factor Graphs
- Belief Networks
- Deep Probabilistic Models: Deep belief networks
- Deep Probabilistic Models: Boltzmann machines
- Sampling in Probabilistic Learning: Gibbs Sampling (Part 1)
- Sampling in Probabilistic Learning: Gibbs Sampling (Part 2)

Distributed and Federated Learning

- Distributed Machine Learning (DML) and its Types
- Distributed Training Architecture
- Frameworks for DML
- Challenges in DML
- Federated Learning (FL)
- Need of Federated Learning
- Architecture of Federated Learning
- Challenges and real-world applications of Federated Learning

Unit 3

Image Generation and Explainable AI

- Computer Vision Foundation (Part 1)
- Computer Vision Foundation (Part 2)
- Automatic Image Captioning
- Image segmentation (Basic)

- Image segmentation (Working)
- Automatic Image Captioning (Working)
- Image generation with Generative adversarial networks
- Object detection
- Attention models for computer vision tasks

Explainable AI

- Explainable AI (XAI) Techniques
- Interpretable ML Models: SHAP
- Interpretable ML Models: LIME
- Attention Mechanisms
- Regulatory Compliance & Trustworthy AI
- Privacy and Security in ML

Practical / Lab Component

Unit 1

- To implement Ridge Regression and Lasso Regression using Python and compare their performance with Linear Regression on a regression dataset.
- To implement a Support Vector Machine (SVM) classifier on a classification dataset and study the effect of different kernels on the classification accuracy using Python
- Implement AGNES (Agglomerative Nesting) and DIANA (Divisive Analysis) for clustering a dataset.
- To implement a Self- Organizing Map (SOM) using Python and apply it for clustering and visualization of high-dimensional data.
- Implement ensemble learning techniques on a real-world dataset and evaluate their performance using cross-validation and analyse the impact of different hyperparameters on model accuracy.

Unit 2

- Implement a Deep Belief Network on a simple dataset and visualize the network learned features.
- To implement Gibbs Sampling in Python and understand how it is used to generate samples from a probability distribution when direct sampling is difficult.
- Implement a Hidden Markov Model (HMM) to predict sequences in time-series data (e.g., weather forecasting, speech recognition).
- Implement a Federated Learning system using a machine learning model.
- To Implement a Federated Learning experiment by dividing one dataset into multiple clients, training models separately, and combining their predictions.

Unit 3

- To implement a simple image segmentation experiment using Python and OpenCV to separate an object or region of interest from an image.
- Implement a basic Generative Adversarial Network (GAN) to generate synthetic images from random noise.
- Implement SHAP and LIME techniques to explain the predictions of a Random Forest Model on a classification task.
- Apply an object detection model to detect and classify objects in an image dataset.
- Case Study: Vehicle Detection in Road Traffic using the concept of Object Detection.

Soft Computing

Course Code: 24CSH-339

Unit 1

Fundamentals of Computing

- Definition of Computing and Computational Models
- Deterministic vs Non-Deterministic Computing
- Hard Computing vs Soft Computing
- Limitations of Traditional Computing Approaches

Chapter 2: Introduction to Soft Computing

- Introduction to Soft Computing
- Major Areas and Components of Soft Computing
- Characteristics of Soft Computing
- Applications in Engineering, AI and Data Science

Chapter 3: Neural Networks

- Introduction and Taxonomy of Neural Networks
- ANN Architecture
- Learning Rules in ANN
- ANN Training Methods
- Perceptron Model
- Multilayer Perceptron Model
- Applications of Neural Networks

Unit 2

Chapter 4: Fuzzy Logic

- Introduction to Fuzzy Logic
- Classical Sets and Fuzzy Sets
- Membership Functions
- Fuzzy Rule Generation
- Operations on Fuzzy Sets
- Aggregation and Combination Operations

Chapter 5: Fuzzy Arithmetic

- Fuzzy Numbers and Linguistic Variables
- Arithmetic Operations on Intervals and Fuzzy Numbers
- Lattice of Fuzzy Numbers
- Fuzzy Equations

Chapter 6: Fuzzy Chapter 6: Fuzzy Systems

- Fuzzy Rule Based Systems
- Predicate Logic and Fuzzy Decision Making
- Fuzzy Control Systems
- Fuzzy Classification
- Min-Max Composition and Defuzzification Methods

Unit 3

Chapter 7: Introduction to Hybrid Systems

- Concept and Need for Hybridization
- Advantages and Types of Hybrid Intelligent Systems
- Neuro-Fuzzy Systems
- Neuro-Genetic Systems
- Fuzzy-Genetic and Neuro-Fuzzy- Genetic Systems

Chapter 8: Neuro Fuzzy Hybrid Systems

- Comparison of Fuzzy Systems and Neural Networks
- Characteristics of Neuro-Fuzzy Hybrid Systems
- Classification of Neuro-Fuzzy Hybrid Systems
- Cooperative Neural Fuzzy Systems
- General Neuro-Fuzzy Hybrid Systems

Chapter 9: Genetic Neuro-Hybrid Systems

- Properties of Genetic Neuro- Hybrid Systems
- Genetic Algorithm Based Back Propagation Network
- Coding, Weight Extraction and Fitness Function
- Reproduction of Offspring and Convergence

Chapter 10: Genetic Algorithms

- Fundamentals, Encoding and Operators of Genetic Algorithms

Practical / Lab Component

Unit 1

- Getting started with Python 3.x and installing TensorFlow, Keras, PyTorch
- Build a fully connected layer architecture using in-built datasets
- Creating an Artificial Neural Network Class: Training the model and applying Sigmoid function

Unit 2

- Program to build Convolutional Neural Network
- Program to implement Multi Layer Perceptron
- Program to implement Feed Forward Network

Unit 3

- Program to implement common operations on Fuzzy Set
- Program to implement Fuzzy Control System: Tipping Problem
- Program to implement Fuzzy Inference System
- Program to implement GANs using Genetic Algorithms
- Handwritten Digit Recognition using Neural Networks
- Medical Diagnosis using Fuzzy Logic System
- Stock Market Prediction using Genetic Algorithms
- Traffic Signal Control using Reinforcement Learning
- Spam Email Classification using Deep Learning

Competitive Coding-II

Course Code: 24CSP-305

Practical / Lab Component

Unit 1

- Contains Duplicate II
- Product of array except self
- Search insert position
- search in rotated sorted array
- Implement Queue using stack
- Largest rectangle in histogram
- Palindrome Linked List
- Odd even Linked List

Unit 2

- Subsets
- Combination Sum
- Lowest Common ancestor of binary tree
- Inorder Successor in BST
- Add Digits
- Find the duplicate Number

Unit 3

- Climbing Stairs
- Frog jump with K distance
- Coin Change
- Longest Common Subsequence
- Number of islands
- Find if path exists in the graph

Advanced Database

Course Code: 24CSP-310

Practical / Lab Component

Unit 1

- Introduction to Advanced DBMS and SQL Commands (DDL, DML, DCL, TCL)
- SQL Joins and Subqueries using Employee Database
- Database Design and Normalization Techniques (1NF to BCNF)
- Transaction Management and ACID Properties
- Views and Query Simplification using Aggregate Functions

Unit 2

- Stored Procedures and Functions in SQL
- Window Functions and Ranking Operations
- Trigger Mechanisms and Transaction Logging
- Query Optimization using Execution Plan Analysis
- Indexing and Performance Tuning Techniques

Unit 3

- Distributed Databases and AWS RDS Introduction
- Introduction to MongoDB and NoSQL Databases
- MongoDB Aggregation Framework and Data Analysis
- Role-Based Access Control and Database Security
- Mini Project Demonstration using SQL and NoSQL Integration

Full Stack -II

Course Code: 24CSP-337

Practical / Lab Component

Unit 1

- Post Composer UI with Platform Constraints
- Draft Management & API Integration
- Redux Toolkit Setup & State Design
- Optimized Selectors & Performance Handling
- JWT-Based Authentication System
- Role-Based Access & Route Protection
- Interactive Calendar for Post Scheduling
- Performance Optimization & Testing

Unit 2

- REST API Design & Implementation
- Global Exception Handling & Logging
- Pagination & Sorting Implementation
- Caching & Query Optimization
- Secure APIs with JWT & RBAC
- Encryption & Secure Credential Storage

Unit 3

- Kafka Producer-Consumer Implementation
- Reliability: Retry, DLQ & Idempotency
- Continuous Integration Pipeline Setup
- Continuous Deployment & Release Strategies
- Full-Stack System Integration
- Capstone Project

Computer Networks

Course Code: 24CST-339

Unit 1

Chapter 1.1

- Fundamentals of Computer Networks
- Computer Networks and Distributed Systems
- Classification of Computer Networks (LAN, MAN, WAN)
- Advantages and Applications of Computer Networks

Chapter 1.2

- Layered Network Architecture and its Need
- Data Communication Components and their Functions
- Representation of Data in Communication Systems
- Data Flow Types: Simplex, Half- Duplex and Full-Duplex
- Network Topologies: Bus, Star, Ring, Mesh and Tree
- Protocols and Networking Standards
- OSI Reference Model and Layer Functions
- Transmission Media: Guided and Unguided Media

Chapter 1.3

- Wired Local Area Network (LAN) Concepts
- Wireless Local Area Network (WLAN) Concepts
- Virtual LAN (VLAN): Concept and Applications

Unit 2

Chapter 2.1

- Virtual LAN (VLAN): Concept and Applications
- Time Division Multiplexing (TDM) for Data Transmission
- Wavelength Division Multiplexing (WDM) and Other Multiplexing Techniques
- Spread Spectrum Techniques for Bandwidth Utilization

Chapter 2.2

- Detection and Error Correction Methods in Data Communication
- Block Coding and Hamming Distance Mechanisms
- Implement Cyclic Redundancy Check (CRC) for Error Detection
- Flow Control and Error Control Protocols (Stop-and-Wait ARQ)
- Go-Back-N ARQ and Selective Repeat ARQ Protocols
- Sliding Window and Piggybacking Techniques
- Random Access and Multiple Access Protocols (ALOHA, Slotted ALOHA)
- CSMA/CD and CSMA/CA Protocols for Medium Access Control

Chapter 2.3

- Switching Techniques and Logical Addressing in Networks
- IPv4 and IPv6 Addressing Schemes
- Address Mapping and Routing Protocols (ARP, RARP, BOOTP, DHCP, Forwarding and Unicast Routing)

Unit 3

Chapter 3.1

- Process-to-Process Communication in the Transport Layer
- Working and Applications of User Datagram Protocol (UDP)
- Features and Operation of Transmission Control Protocol (TCP)
- TCP, UDP, and SCTP Transport Layer Protocols
- Congestion Control Mechanisms in SCTP
- Quality of Service (QoS) Requirements in Computer Networks
- Leaky Bucket and Token Bucket Algorithms for QoS Improvement

Chapter 3.2

- Domain Name System (DNS) and Dynamic DNS (DDNS) Services
- Application Layer Protocols: TELNET, EMAIL, and FTP
- WWW and HTTP Protocols for Web Communication
- SNMP and Bluetooth Protocols in Network Applications
- Firewalls in Network Protection

Chapter 3.3

- Electronic Mail Services, Directory Services, and Network Management Functions
- Cryptographic Techniques for Secure Communication
- Network Security Mechanisms in Modern Networks

Predictive analytics

Course Code: 24CST-349

Unit 1

Introduction to Predictive Analysis

- Introduction to Predictive Analysis and its significance
- The role of predictive analysis in decision-making
- Characteristics of data for predictive analysis
- Predictive Analysis Process: Data collection, data preprocessing, model selection, model training, model evaluation
- Ethical considerations in predictive analysis

Data Preparation for Predictive Analysis

- Data cleaning and preprocessing: Handling missing values, outliers, and noise
- Data transformation: Feature scaling, encoding categorical variables, and feature engineering
- Data splitting: Train-test split and cross-validation
- Exploratory Data Analysis (EDA) techniques
- Handling imbalanced datasets

Predictive Modelling Techniques

- Supervised learning algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting
- Model evaluation metrics: Mean Squared Error (MSE), R-squared, Accuracy, Precision, Recall, F1- score
- Hyperparameter tuning and model selection
- Overfitting and underfitting: Bias- variance trade-off
- Model interpretation and visualization

Unit 2

Data Storage and Management for Predictive Analysis

- Data storage considerations: Data warehouses, databases, and cloud-based storage
- Data extraction, transformation, and loading (ETL) processes
- Data encryption and security in predictive analysis
- Managing data storage performance

Feature Engineering

- Understanding feature importance
- Feature selection techniques: Filter, Wrapper, and Embedded methods
- Feature scaling and normalization
- Handling time-series data and categorical features
- Feature extraction and dimensionality reduction

Data Visualization for Predictive Analysis

- Importance of data visualization in predictive analysis
- Types of data visualizations: scatter plots, bar charts, histograms, heatmaps, etc.
- Data visualization libraries and tools (e.g., Matplotlib, Seaborn, Tableau)
- Visualizing feature distributions and relationships
- Visualizing model performance and results
- Interactive and dynamic data visualizations

Unit 3

Advanced Predictive Analysis Techniques

- Time series analysis and forecasting
- Ensemble methods: Bagging and Boosting
- Deep learning for predictive analysis: Introduction to neural networks and deep learning frameworks
- Model interpretability and explainability
- Handling textual data for sentiment analysis

Ethical and Privacy Considerations in Predictive Analysis

- Data privacy laws and regulations (e.g., GDPR, CCPA)
- Ethical issues in predictive analysis: Bias, fairness, and transparency
- Privacy-preserving predictive analysis techniques
- Responsible data use and AI ethics
- Bias and fairness in predictive analysis algorithms

Anomaly Detection and Unsupervised Learning

- Anomaly detection: Types of anomalies (point anomalies, contextual anomalies, collective anomalies)
- Anomaly detection algorithms: Isolation Forest, One-Class SVM, DBSCAN
- Applications of anomaly detection
- Fraud detection, network security, fault detection
- Evaluation metrics for anomaly detection

Soft Skills-III

Course Code: 24TDP-311

Practical / Lab Component

Unit 1

- Exponential Thinking, Resume to JD Connect
- Unscripted Monologue, Basics of Psychometric Test
- Power Words, Pitch the Impossible
- Crisis Communication & Rapid Response, Embracing Professionalism
- Cover Letter Writing, Making Commercials

Unit 2

- Event Simulations, Grooming Etiquette
- Elevator Pitch, Idea Generation Skills - 4 (Attribute Analysis)
- Case Study Based GD, Critical Reasoning
- Negotiation Skills, Conditional Sentences
- Present Your Ideas, Personal Branding

Unit 3

- Visume Skills
- STAR Interview
- Essay Writing
- Turn Around the Tables
- Behavioural Interview Skills
- Encouraging Innovation
- Convergent and Divergent Thinking
- Figures of Speech
- Decision-Making Skills
- Word Analogy

Aptitude-III

Course Code: 24TDT-312

Unit 1

LCM and HCF

- Introduction to LCM and HCF, Prime Factorization Method to find LCM and HCF.
- Problems based on same and different remainders, Relation b/w LCM & HCF, HCF and LCM of Fractions.

Time, Speed and Distance

- Relation between time, speed and distance, Conversion of speed from m/s to km/h and vice versa, Concept of proportionality Concept of distance constant, Concept of Time Constant
- Concept of relative speed, Concept of Average speed

Problems on Trains

- Concept of point Train (Point Object, Length Object, Relative Speed), Problems related to cross a pole, train, bridge and platform
- Concept of Relative Speed, Two trains coming from two ends and meeting at a point

Boats and Stream

- Introduction about Topic, Swimmer/boat to travel upstream and downstream

Linear and Circular Races

- Problems with races on linear tracks: head start and dead heat finish, head start and early finish, head start and late finish , Problems on circular track: first meeting point, meeting at starting point and number of meetings

Revision unit- 1

- Practice LCM & HCF, Time Speed Distance
- Practice Problems on Trains, Linear & Circular Races

Unit 2

Calendar

- Introduction about Topic, Concept of odd days, Concept of Leap Year
- Concept of Non-Leap years and Centuries, Repetition of Calendar

Time and Work

- Introduction about Topic, Concept of Time and Work, Relationship b/w Time, Work & Efficiency, LCM approach
- Practice Questions based on Remaining work, Alternate days, Efficiency, Practice Wages & negative efficiency etc.

Chain Rule

- Introduction about Topic, Concept of Direct Proportional & Indirect Proportional, Problem based on Chain rule

Pipes & Cistern

- Basic Concepts, Rules and Tricks with examples, Efficiency, Inlet outlet pipes, Alternate, Capacity based questions

Permutation and Combination

- Concept of Permutation and Combination, Formulas of Permutation and Combination, Fundamental Principle of Counting
- Problems related to arrangement of letters and numbers with or without Repetition, Problems related to lines , diagonals, handshakes and circular permutation

Revision unit- 2

- Practice Calendar, Time & Work
- Practice Pipes & Cistern, Permutation & Combination

Unit 3

Sequence and Series

- Introduction to sequence and Series, Arithmetic Progression, Arithmetic Mean (AM), Geometric Progression, Geometric Mean (GM)
- Harmonic Progression, Harmonic Mean (HM), Relation between AM, GM and HM

Surface Area (Mensuration - 3D)

- Introduction about Topic, Concept of 3D figures (types and characteristics): Cube, Cuboid, Cylinder, Cone, Sphere and Hemisphere
- Concept of lateral/curved and Total Surface Area

Volume (Mensuration - 3D)

- Recalling of Previous Topic, Introduction about Topic
- Volume of 3D figures, Practice questions & Data Sufficiency based questions

Seating Arrangement (Square and Rectangle Based)

- Introduction about Topic, Concept of square seating arrangement, Question based on square table arrangement
- Concept of rectangle arrangement, Question based on rectangle table arrangement

Revision unit- 3

- Practice Sequence & Series, Surface Area
- Practice Volume, Seating Arrangement(Square and Rectangle)

Entrepreneurship

Course Code: 24UCT-399

Unit 1

Entrepreneur ship Foundations

- Use an Entrepreneurial Mindset to Find Success and Fulfillment at Work
- Introduction to new venture
- Employee to Entrepreneur
- Become an Entrepreneur Inside a Company
- Unique Ways to Generate Creative Ideas

Unit 2

Business plan development

- New product development
- Product development process
- Creating a Business Plan
- Revenue, Staffing, and Expense Models Explained
- Revenue recognition

Unit 3

Vendor Management

- Decks, pitching, and fundraising
- Social Selling: Reaching Prospects
- Advertising, Business growth ,Social effects
- Project Management: Government Projects